

openGUTS Report

Project:

GUTS_RF_Limnomysis_81_Noprefitbackgroundhazard

Project file:

No project file saved or loaded

Project description (optional):

No project description available

Software version:

openGUTS - 1.2

Date of report creation:

23/07/2026 11:41:13

Calibration

Calibration input data

Data set 1

File: Data edited manually

Description (optional):

Control group: 'Control'

Survival data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	80	20	20	20	20	20	20
0.042	79	20	20	20	19	20	16
0.083	79	20	20	20	19	17	9
0.16	79	20	20	19	19	14	0
0.33	77	19	20	19	19	12	0
1	72	18	19	19	18	7	0
1.208	72	18	19	19	15	7	0
2	66	18	18	18	14	4	0
3	62	16	16	16	11	3	0
4	57	14	15	16	8	2	0

Concentration data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	0	3	10	30	100	300	1000
0.2	0	3	9.9	29.7	99	296.9	989.5
0.4	0	2.9	9.8	29.4	97.9	293.7	979.1
0.6	0	2.9	9.7	29.1	96.9	290.7	968.9
0.8	0	2.9	9.6	28.8	95.9	287.6	958.7
1	0	2.8	9.5	28.5	94.9	284.6	948.7
1.2	0	2.8	9.4	28.2	93.9	281.6	938.7
1.4	0	2.8	9.3	27.9	92.9	278.7	928.9
1.6	0	2.8	9.2	27.6	91.9	275.7	919.2

1.8	0	2.7	9.1	27.3	91	272.9	909.5
2	0	2.7	9	27	90	270	900
2.2	0	2.7	8.9	26.7	89.1	267.2	890.5
2.4	0	2.6	8.8	26.4	88.1	264.4	881.2
2.6	0	2.6	8.7	26.2	87.2	261.6	872
2.8	0	2.6	8.6	25.9	86.3	258.9	862.8
3	0	2.6	8.5	25.6	85.4	256.1	853.8
3.2	0	2.5	8.4	25.3	84.5	253.5	844.8
3.4	0	2.5	8.4	25.1	83.6	250.8	836
3.6	0	2.5	8.3	24.8	82.7	248.2	827.2
3.8	0	2.5	8.2	24.6	81.9	245.6	818.5
4	0	2.4	8.1	24.3	81	243	810

Calibration settings

Calibration parameter settings for GUTS-RED-SD:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	1096	Log
mw	Yes	0.0007399	990	Norm
hb	Yes	1E-6	0.15	Norm
bw	Yes	2.634E-5	334.5	Log
Fs	No	1	1	Norm

Calibration parameter settings for GUTS-RED-IT:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	80.23	Log
mw	Yes	0.0007399	1212	Log
hb	Yes	1E-6	0.15	Norm
bw	No	Inf	Inf	Norm
Fs	Yes	1.05	20	Log

Note:

Expert settings were used (calibration settings may have been changed).

Parameter ranges were changed manually.

Calibration results

Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 52.99 (31.22 - 203.7)

mw: 246 (219.2 - 267.3)

hb: 0.09348 (0.06989 - 0.1219)

bw: 0.02289 (0.01407 - 0.03805)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.9563

Normalised root-means-square error (NRMSE): 6.539 %

Minus log-likelihood (MLL): 261.08

AIC: 530.17

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	2.447 %
1	T1	1.197 %
1	T2	6.197 %
1	T3	11.2 %
1	T4	-28.8 %
1	T5	2.468 %
1	T6	-8.09E-25 %

GUTS-RED-SD results table for LC_{x,t} [[C]], with 95% CI:

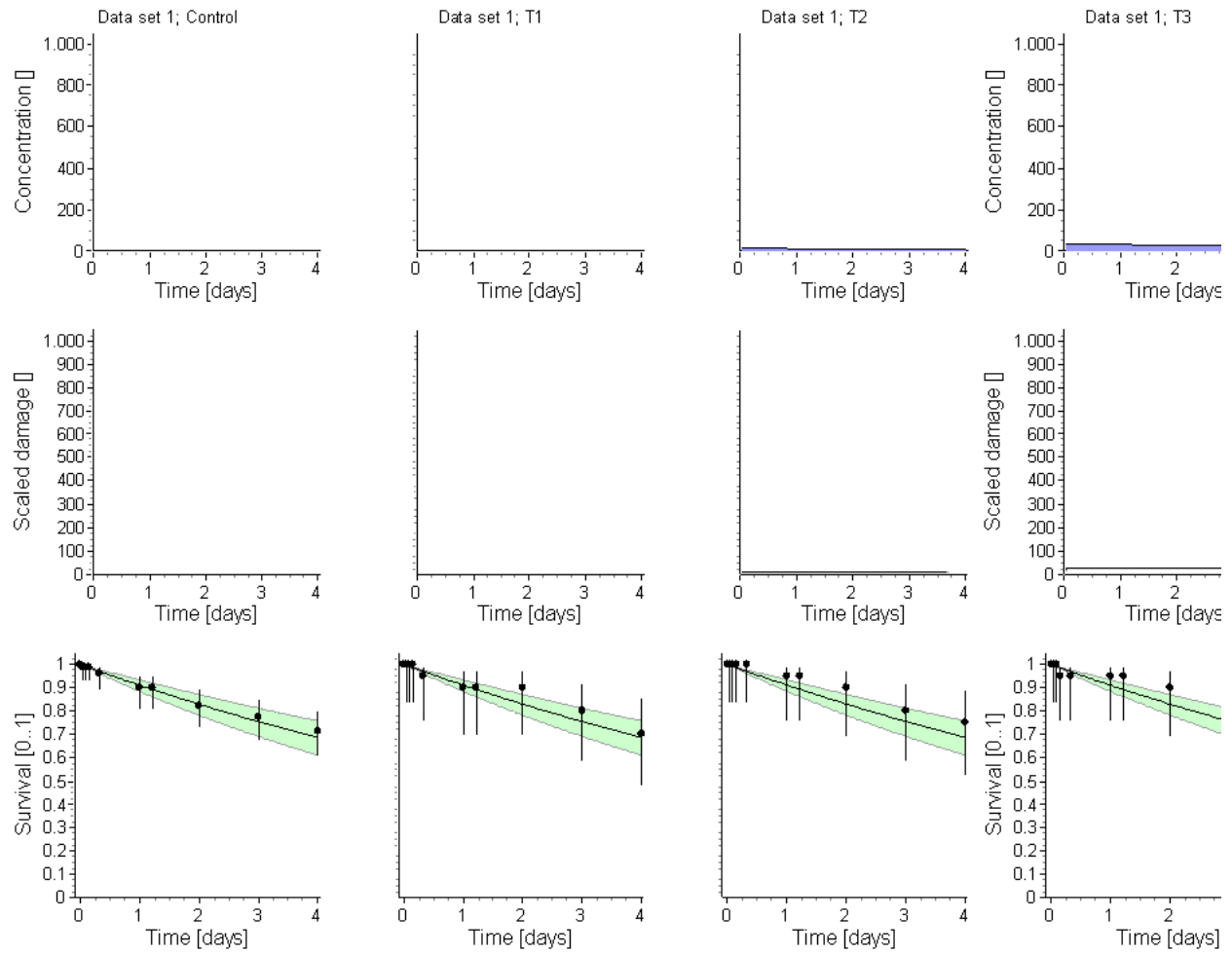
Time [d]	LC50	LC20	LC10
1	278.2 (259.1 - 296.5)	256.6 (232.4 - 275.8)	251.1 (224.8 - 271.6)
2	261.7 (239.5 - 280)	251.1 (224.9 - 271.6)	248.4 (221.2 - 269.6)
3	256.4 (232.2 - 275.6)	249.4 (222.5 - 270.3)	247.6 (220 - 268.9)

4	253.7 (228.5 - 273.6)	248.5 (221.3 - 269.6)	247.2 (219.5 - 268.6)
7	250.4 (223.9 - 271)	247.4 (219.8 - 268.8)	246.7 (218.7 - 268.2)
10	249.1 (222.1 - 270)	247 (219.2 - 268.5)	246.5 (218.5 - 268.1)
14	248.2 (220.8 - 269.4)	246.7 (218.8 - 268.3)	246.3 (218.3 - 268)
21	247.4 (219.8 - 268.8)	246.5 (218.5 - 268.1)	246.2 (218.1 - 267.9)
27	247.1 (219.4 - 268.6)	246.4 (218.3 - 268)	246.2 (218.1 - 267.9)
28	247.1 (219.3 - 268.6)	246.3 (218.3 - 268)	246.2 (218 - 267.9)
42	246.7 (218.8 - 268.3)	246.2 (218.1 - 267.9)	246.1 (218 - 267.8)
50	246.6 (218.7 - 268.2)	246.2 (218.1 - 267.9)	246.1 (217.9 - 267.8)
100	246.3 (218.2 - 268)	246.1 (218 - 267.8)	246 (217.9 - 267.8)
182	246.2 (218 - 267.9)	246.1 (217.9 - 267.8)	246 (217.9 - 267.7)

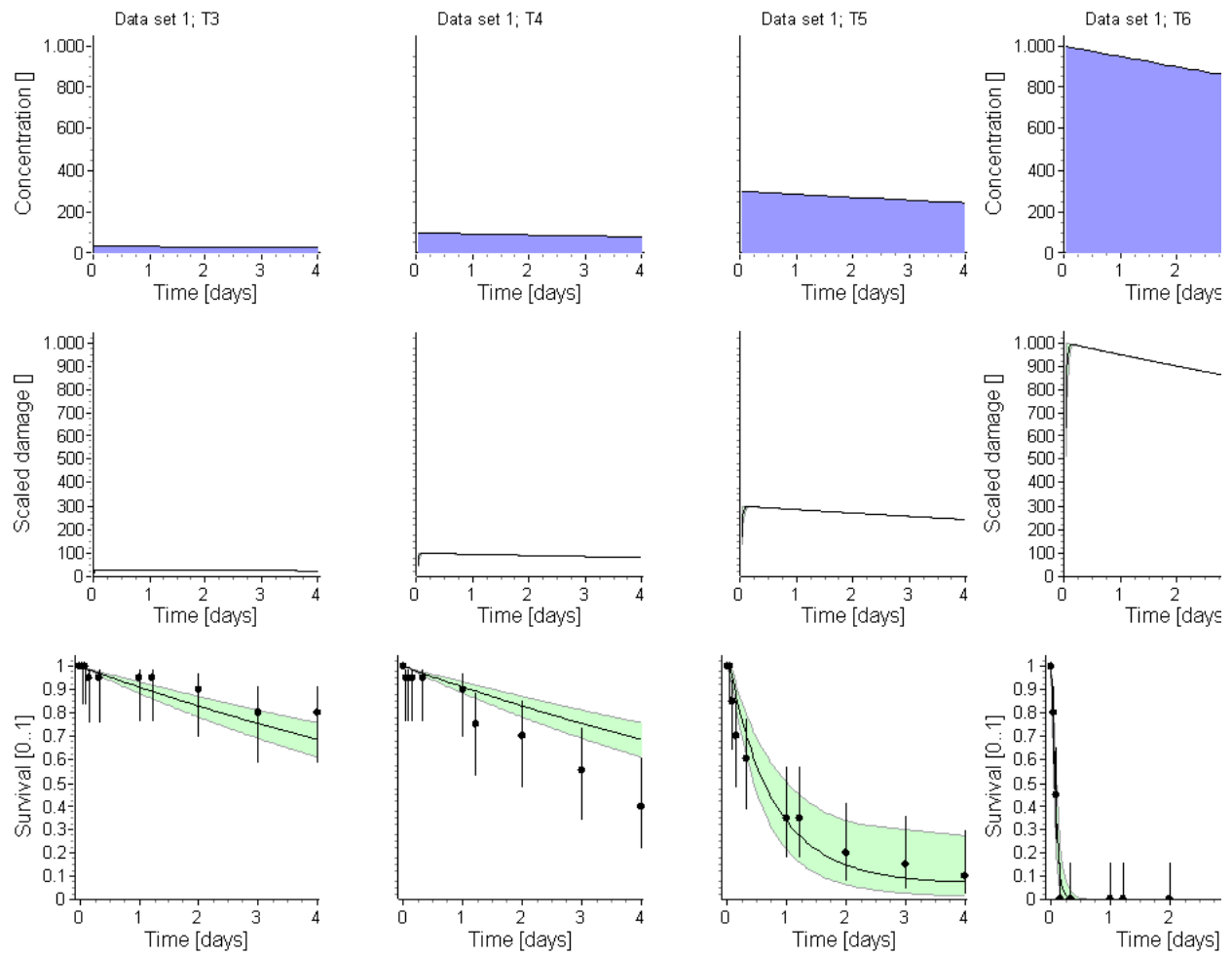
Plots for GUTS-RED-SD calibration:

Parameter space plot for the calibration of GUTS-RED-SD:

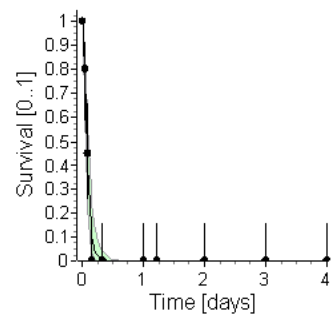
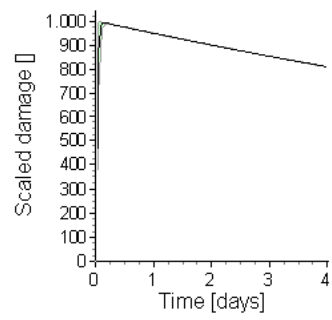
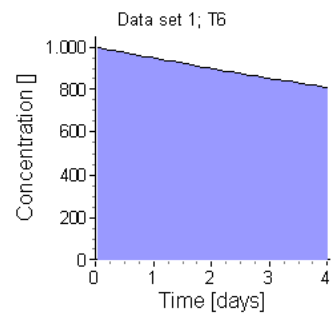
Exposure, damage and survival plots for the calibration of GUTS-RED-SD:



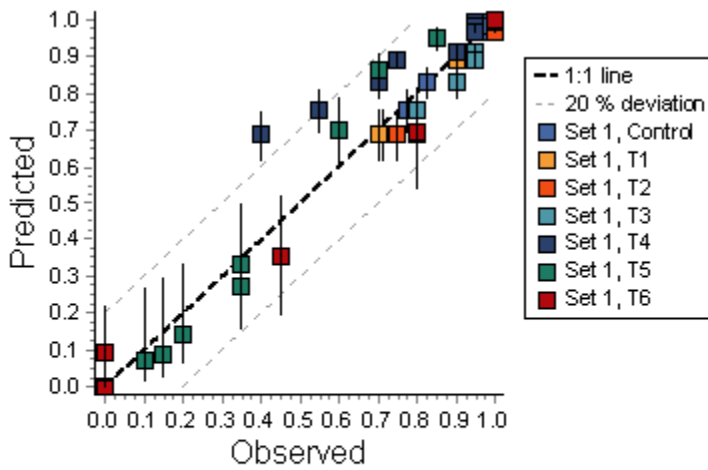
... continued plot:



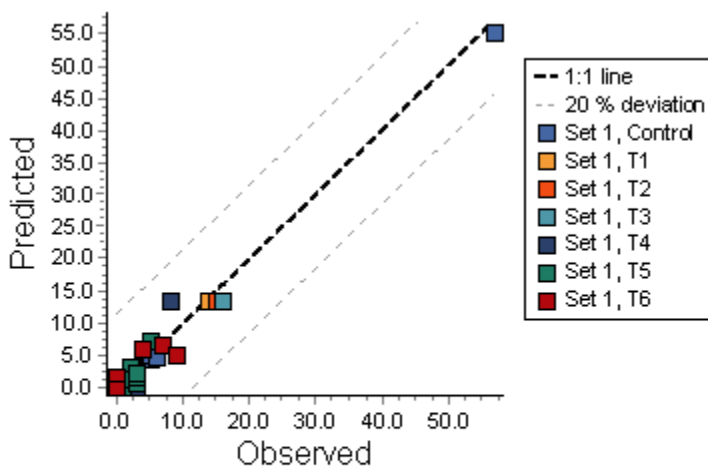
... continued plot:



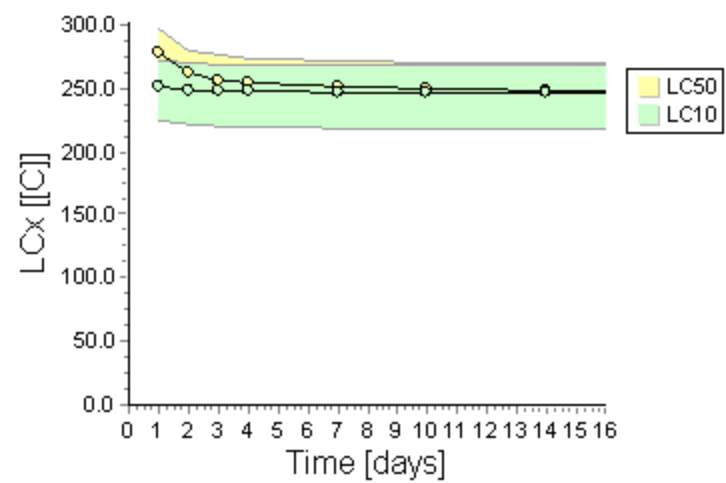
Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):



Fitted parameters for GUTS-RED-IT:

Best fit parameter values and their 95% CI

kd: 4.369 (1.595 - 6.72)

mw: 254.7 (129 - 346.2)

hb: 0.1006 (0.07612 - 0.1298)

Fs: 3.227 (2.31 - 5.768)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-IT):

Model efficiency (NSE, r-square): 0.9568

Normalised root-means-square error (NRMSE): 6.833 %

Minus log-likelihood (MLL): 265.04

AIC: 538.08

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	4.372 %
1	T1	3.122 %
1	T2	8.124 %
1	T3	13.19 %
1	T4	-24 %
1	T5	-17.8 %
1	T6	-1.08 %

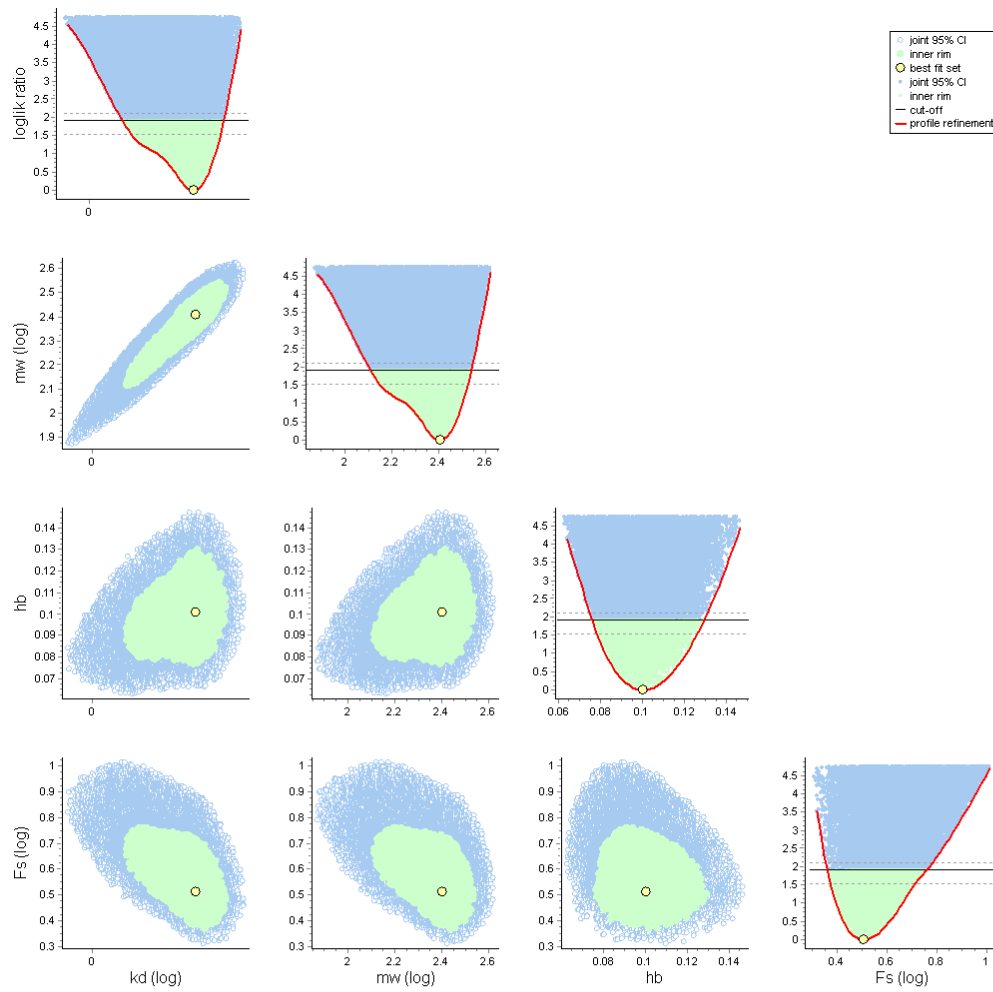
GUTS-RED-IT results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	258 (155.3 - 351)	165.6 (84.17 - 231.2)	127.8 (58.15 - 184.6)
2	254.7 (131.3 - 350.3)	163.5 (70.94 - 230.7)	126.2 (49.01 - 184.2)
3	254.7 (126.9 - 350.3)	163.5 (68.92 - 230.7)	126.2 (47.61 - 184.2)
4	254.7 (126 - 350.3)	163.5 (68.47 - 230.7)	126.2 (47.36 - 184.2)
7	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)

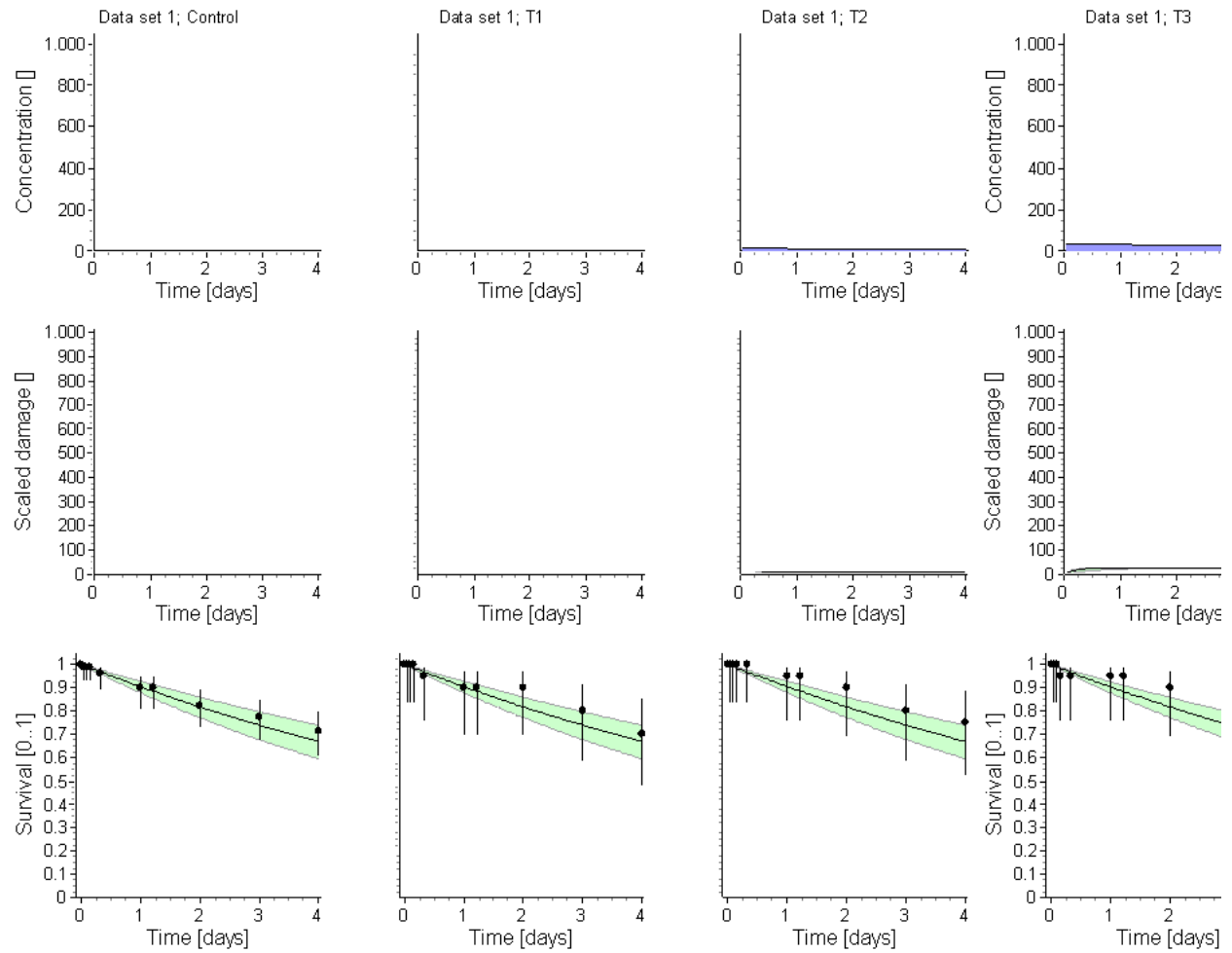
10	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
14	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
21	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
27	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
28	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
42	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
50	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
100	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)
182	254.7 (125.8 - 350.3)	163.5 (68.34 - 230.7)	126.2 (47.3 - 184.2)

Plots for GUTS-RED-IT calibration:

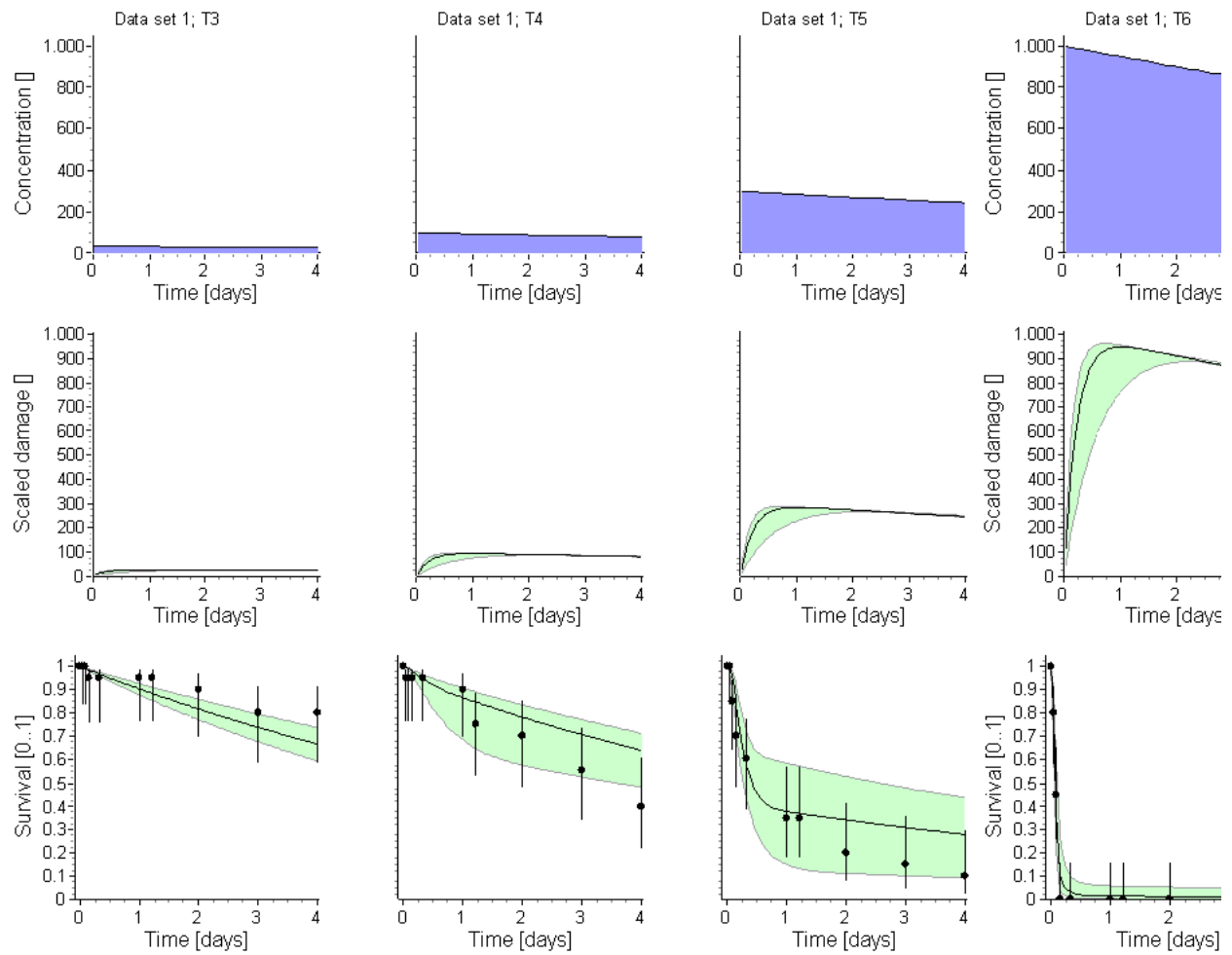
Parameter space plot for the calibration of GUTS-RED-IT:



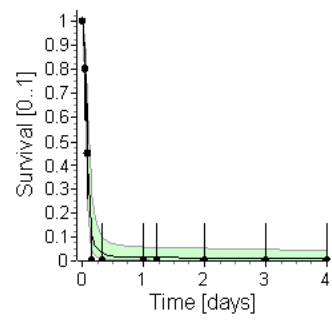
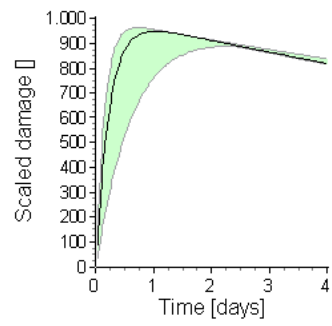
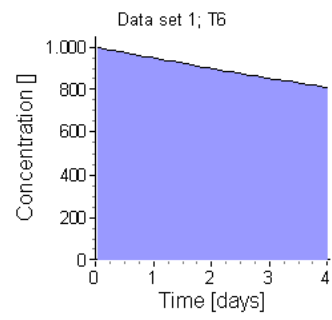
Exposure, damage and survival plots for the calibration of GUTS-RED-IT:



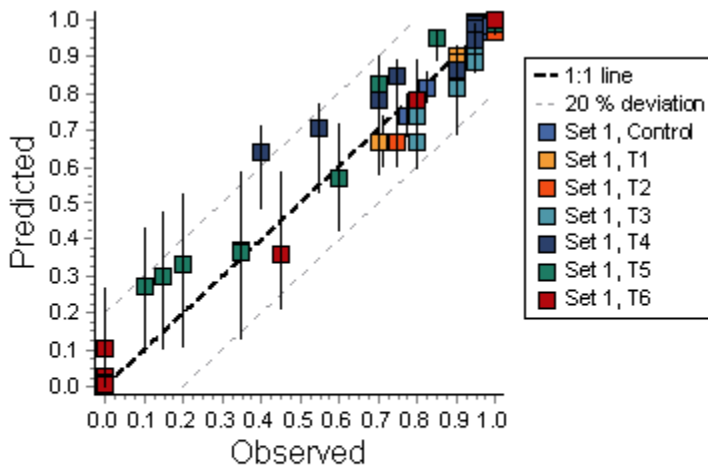
... continued plot:



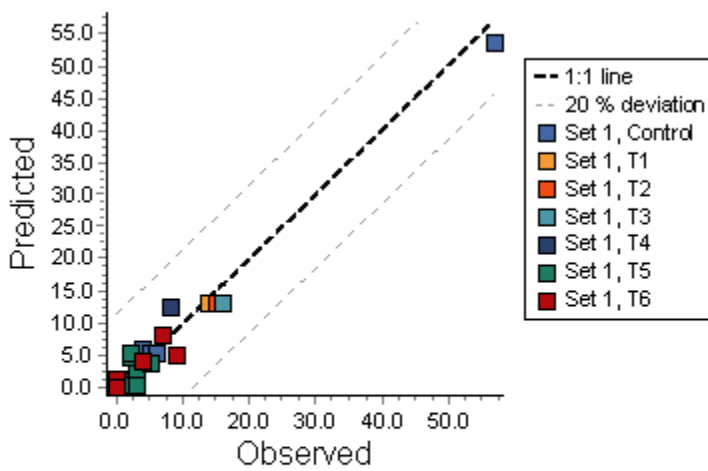
... continued plot:



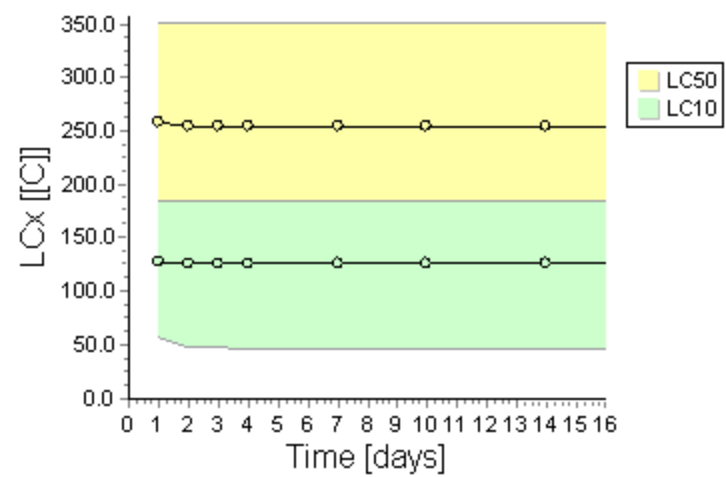
Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):



Validation

No validation performed!

Predictions

No predictions performed!